Trigonometric Identities (TE1)

Math 172 lecture notes · Section 4.4.1

Learning objective: use the sum and difference, double-angle, and power-reducing formulas for cosine, sine, and tangent, and the Pythagorean identities.

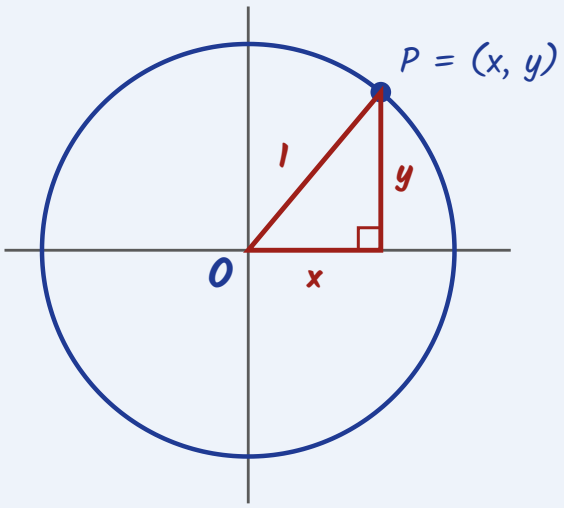
Try each part before you open the answer. Your thinking is what makes it stick.

Problem from your notes Text from the handout Extra note (not in your handout)

Overview

FROM THE HANDOUT

Pythagorean Identities, as we have seen in a previous section, are derived from the Pythagorean Theorem ($a^2 + b^2 = c^2$). For example, consider a point P on the unit circle, with coordinates (x, y). If we draw a right triangle (as shown in the figure below), the Pythagorean Theorem says that $x^2 + y^2 = 1$.



But, remember, that the x-coordinate of the point corresponds to cos θ and the y-coordinate corresponds to sin θ. Thus, we get: **$\sin^2\theta + \cos^2\theta = 1$**. Pythagorean Identities are used in solving many trigonometric problems where one trigonometric ratio is given and we are expected to find the other trigonometric ratios. The next two activities will lead us to find the other two Pythagorean Identities.

Activity 1

Activity 1 · from your notes

Let's begin with the Pythagorean Identity, $\cos^2\theta + \sin^2\theta = 1$.

(a) If we divide each term by $\cos^2\theta$, what would the resulting equation be?

Answer (a)

$$\frac{\cos^2\theta}{\cos^2\theta} + \frac{\sin^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta}$$

Domain check: we can only divide by $\cos^2\theta$ when $\cos\theta \neq 0$, so this identity holds wherever tan θ and sec θ are defined.

(b) Recall $\csc\theta = \frac{1}{\sin\theta}$, $\sec\theta = \frac{1}{\cos\theta}$, $\cot\theta = \frac{\cos\theta}{\sin\theta}$. Rewrite each fractional term in your equation from part (a) as a single trigonometric function.

> **Hint:** Write $\frac{\sin^2\theta}{\cos^2\theta}$ as $\left(\frac{\sin\theta}{\cos\theta}\right)^2$. Which function is sin over cos?

Answer (b)

$\frac{\cos^2\theta}{\cos^2\theta} = 1$, $\frac{\sin^2\theta}{\cos^2\theta} = \left(\frac{\sin\theta}{\cos\theta}\right)^2 = \tan^2\theta$, $\frac{1}{\cos^2\theta} = \sec^2\theta$

$$1 + \tan^2\theta = \sec^2\theta$$

Note

FROM THE HANDOUT

In the previous example, we derived the second Pythagorean Identity: $1 + \tan^2\theta = \sec^2\theta$

Activity 2

Activity 2 · from your notes

We can also derive the third Pythagorean identity from $\sin^2\theta + \cos^2\theta = 1$.

(a) What do we get if we divide each term by $\sin^2\theta$?

Answer (a)

$$\frac{\sin^2\theta}{\sin^2\theta} + \frac{\cos^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta}$$

> **Domain check:** dividing by $\sin^2\theta$ requires $\sin\theta \neq 0$, so this identity holds wherever $\cot\theta$ and $\csc\theta$ are defined.

(b) Rewrite each fractional term in the equation you got in part (a) as a single trigonometric function.

Answer (b)

$$1 + \cot^2\theta = \csc^2\theta$$

Note

FROM THE HANDOUT

In the previous example, we derived the third Pythagorean Identity: $1 + \cot^2\theta = \csc^2\theta$

Theorem

FROM THE HANDOUT

The three ***Pythagorean Identities*** *are:*

- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \csc^2\theta$

Note that each of these identities can be written in different forms, which are all equivalent to one another. For example, $\sin^2\theta + \cos^2\theta = 1$ *can be rewritten as* $1 - \sin^2\theta = \cos^2\theta$ *or* $1 - \cos^2\theta = \sin^2\theta$*. You will want to become familiar with all these forms.*

Note

FROM THE HANDOUT

Other identities and formulas (in addition to the Pythagorean Identities) can be used to solve various mathematical problems. The next few activities will lead us through an exploration of other types of identities and formulas.

Activity 3

Activity 3 · from your notes

Refer back to the unit circle to determine the exact value of $\sin\left(\pi + \frac{\pi}{2}\right)$**.**

(a) *What is the value of* $\sin\pi$*?*

Answer (a)

$\sin\pi = 0$

(b) What is the value of $\sin(\frac{\pi}{2})$?

Answer (b)

$$\sin\left(\frac{\pi}{2}\right) = 1$$

(c) Based on your answers from parts (a) and (b), what do you think the value of $\sin(\pi + \frac{\pi}{2})$ is?

Answer (c)

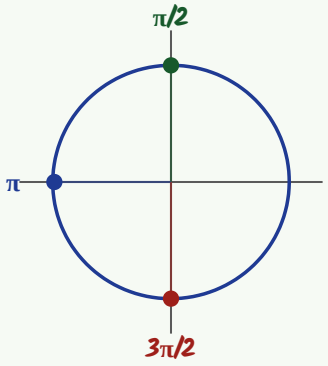
A natural guess is to add them: $\sin \pi + \sin \frac{\pi}{2} = 0 + 1 = 1$. Keep that guess in mind and test it.

(d) Let's test your conjecture from part (c). What is the value of $\pi + \frac{\pi}{2}$?

Answer (d)

$$\pi + \frac{\pi}{2} = \frac{3\pi}{2}$$

And $\sin(\frac{3\pi}{2}) = -1$: the bottom of the unit circle.



(e) Is it safe to assume that $\sin(\pi + \frac{\pi}{2}) = \sin \pi + \sin(\frac{\pi}{2})$?

Answer (e)

The actual value is $\sin(\frac{3\pi}{2}) = -1$, but adding gave 1.

No: $\sin(\alpha + \beta)$ is not $\sin \alpha + \sin \beta$

Note

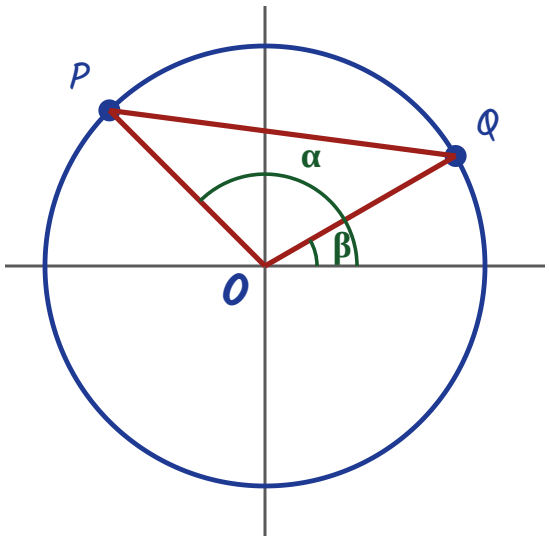
FROM THE HANDOUT

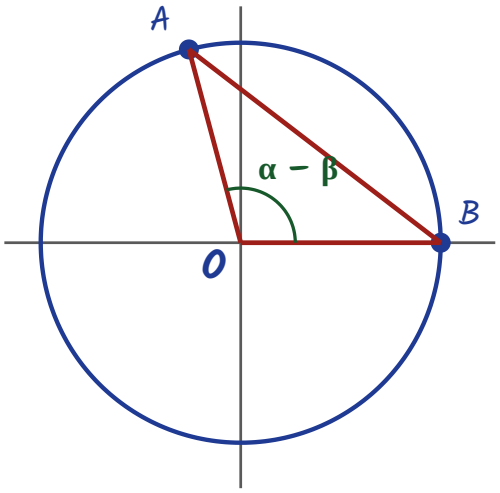
Notice that in the previous example, we saw that $\sin(\pi + \frac{\pi}{2}) \neq \sin \pi + \sin \frac{\pi}{2}$. This is also true for both cosine and tangent as well; that is, if we want to find the cosine or tangent of the sum of two angles, we cannot assume that it is equal to the sum of the two trigonometric functions of each angle. For example, $\cos(\pi + \frac{\pi}{2}) \neq \cos \pi + \cos \frac{\pi}{2}$ and $\tan(\pi + \frac{\pi}{2}) \neq \tan \pi + \tan \frac{\pi}{2}$. The same is true for finding the difference of two angles.

Activity 4

Activity 4 · from your notes

Recall that the coordinates of points on the unit circle are given by $(\cos \theta, \sin \theta)$. In the first unit circle shown below, point P makes an angle α with the positive x-axis and has coordinates $(\cos \alpha, \sin \alpha)$, and point Q makes an angle β with the positive x-axis and has coordinates $(\cos \beta, \sin \beta)$. In the second figure, the triangle is rotated so that point B has coordinates $(1, 0)$.





(a) Note that angle AOB is equal to $(\alpha - \beta)$. What are the coordinates of point A?

Answer (a)

A is on the unit circle at angle $\alpha - \beta$.

$$A = (\cos(\alpha - \beta), \sin(\alpha - \beta))$$

(b) Triangles POQ and AOB are rotations of one another. What can we say about the lengths of PQ and AB?

Answer (b)

Rotating a triangle doesn't change its side lengths.

$$PQ = AB$$

(c) Use the distance formula, $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$, to find the length of PQ. What do you get when you plug in the coordinates of point P and point Q?

> **Hint:** Point P has coordinates $(\cos\alpha, \sin\alpha)$ and point Q has coordinates $(\cos\beta, \sin\beta)$.

Answer (c)

$$d = \sqrt{((\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2)}$$

Subtracting in the other order, $(\cos\beta - \cos\alpha)^2$ and so on, gives the same result once squared.

(d) Begin simplifying by applying $(a - b)^2 = a^2 - 2ab + b^2$. What do you get when squaring the two binomials under the radical?

Answer (d)

$$d = \sqrt{(\cos^2\alpha - 2\cos\alpha\cos\beta + \cos^2\beta + \sin^2\alpha - 2\sin\alpha\sin\beta + \sin^2\beta)}$$

(e) Simplify further by applying the Pythagorean Identity $\cos^2\theta + \sin^2\theta = 1$. What does your answer from part (d) simplify to?

> **Hint:** You may need to rearrange some of your terms before applying the Pythagorean Identity.

Answer (e)

Regroup: $(\cos^2\alpha + \sin^2\alpha) + (\cos^2\beta + \sin^2\beta) = 1 + 1 = 2$.

$$d = \sqrt{(2 - 2\cos\alpha\cos\beta - 2\sin\alpha\sin\beta)}$$

(f) Using the same steps as in parts (c)–(e), use the distance formula to find the distance between points A and B.

> **Hint:** Point A has coordinates $(\cos(\alpha - \beta), \sin(\alpha - \beta))$ and point B has coordinates $(1, 0)$.

Answer (f)

$AB = \sqrt{((\cos(\alpha - \beta) - 1)^2 + (\sin(\alpha - \beta) - 0)^2)}$

$= \sqrt{(\cos^2(\alpha - \beta) - 2\cos(\alpha - \beta) + 1 + \sin^2(\alpha - \beta))} = \sqrt{(1 - 2\cos(\alpha - \beta) + 1)}$

$$\boxed{d = \sqrt{(2 - 2\cos(\alpha - \beta))}}$$

(g) From part (b), $PQ = AB$, so $\sqrt{(2 - 2\cos\alpha\cos\beta - 2\sin\alpha\sin\beta)} = \sqrt{(2 - 2\cos(\alpha - \beta))}$. Square both sides and solve for $\cos(\alpha - \beta)$.

Answer (g)

$2 - 2\cos\alpha\cos\beta - 2\sin\alpha\sin\beta = 2 - 2\cos(\alpha - \beta)$

Subtract 2 from both sides, then divide by -2:

$$\boxed{\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta}$$

Note

FROM THE HANDOUT

In the previous example, we derived the difference formula for cosine. The development of the sum/difference formulas are similar for sine and cosine and same for $\tan(\alpha - \beta)$ compared to $\tan(\alpha + \beta)$.

Theorem

FROM THE HANDOUT

The **Sum and Difference Formulas** in trigonometry are used to find the value of the trigonometric functions at specific angles where it is easier to express the angle as the sum or difference of unique angles such as 0°, 30°, 45°, 60°, 90°, and 180°. The six main trigonometric sum and difference formulas are:

- $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$
- $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$
- $\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

- $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$
- $\tan(\alpha + \beta) = \dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$
- $\tan(\alpha - \beta) = \dfrac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$

Sign check

✎ EXTRA NOTE: NOT IN YOUR HANDOUT

Notice the signs flip on the cosine formulas: a **sum** inside cos gives a **minus** in the middle, and a difference gives a plus. Sine keeps the same sign.

Activity 5

Activity 5 · from your notes

Use the unit circle and a Sum or Difference Formula to find the exact value of cos 75°.

(a) Split 75° into the sum of two angles which can be found on the unit circle (use unique angles such as 0°, 30°, 45°, 60°, 90°, and 180°, etc.).

Answer (a)

$75° = 45° + 30°$

(b) Rewrite cos 75° as $\cos(\alpha + \beta)$, where α and β are the two angles you found in part (a).

Answer (b)

$\cos 75° = \cos(45° + 30°)$

(c) Apply a Sum Formula to rewrite cos 75° with four trigonometric functions.

Hint: Sum inside cosine means minus in the middle.

Answer (c)

$\cos 75° = \cos 45°\cos 30° - \sin 45°\sin 30°$

(d) Use the unit circle to find the value of each trigonometric function in the formula you wrote in part (c).

Answer (d)

$$\cos 75° = \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$$

(e) Simplify your answer from part (d).

Answer (e)

$$\frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4}$$

$$\cos 75° = \frac{\sqrt{6} - \sqrt{2}}{4}$$

Check: $(\sqrt{6} - \sqrt{2}) \div 4 \approx 0.2588$, and a calculator gives $\cos 75° \approx 0.2588$ ✓

Activity 6

Activity 6 · from your notes

Determine whether the statement is true or false by referring to previous work: $\sin(45° - 30°) = \sin 45° - \sin 30°$.

True or false?

Hint: Work out each side separately. Do they match?

Answer

Left side: $\sin 15° \approx 0.2588$. Right side: $\frac{\sqrt{2}}{2} - \frac{1}{2} \approx 0.7071 - 0.5 = 0.2071$.

False

Just like Activity 3: the sine of a difference is not the difference of the sines.

Activity 7

Activity 7 · from your notes

Apply a Sum or Difference Formula to find the exact value of sin 165°.

Try it, then check.

> **Hint:** *Look for two unit-circle angles that add (or subtract) to 165°, such as 120° + 45° or 135° + 30°.*

Answer

$165° = 120° + 45°$, *so use the sum formula for sine:*

$\sin 165° = \sin 120° \cos 45° + \cos 120° \sin 45°$

$= \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(-\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4}$

$$\sin 165° = \frac{\sqrt{6} - \sqrt{2}}{4}$$

Same value as cos 75°. That makes sense: sin 165° = sin 15° = cos 75°.

Activity 8

Activity 8 · from your notes

Suppose $\alpha = \beta$ in each sum formula.

(a) *Rewrite both sides of the sum formula for sine, $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$, in terms of α.*

Answer (a)

$\sin(\alpha + \alpha) = \sin\alpha\cos\alpha + \cos\alpha\sin\alpha$

$$\sin(2\alpha) = 2\sin\alpha\cos\alpha$$

(b) *Rewrite both sides of the sum formula for cosine, $\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$, in terms of α.*

Answer (b)

$\cos(\alpha + \alpha) = \cos\alpha\cos\alpha - \sin\alpha\sin\alpha$

$$\cos(2\alpha) = \cos^2\alpha - \sin^2\alpha$$

(c) Rewrite both sides of the sum formula for tangent, $\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$, in terms of α.

Answer (c)

$$\tan(\alpha + \alpha) = \frac{\tan\alpha + \tan\alpha}{1 - \tan\alpha\tan\alpha}$$

$$\boxed{\tan(2\alpha) = \frac{2\tan\alpha}{1 - \tan^2\alpha}}$$

Domain check: the right side needs $\cos\alpha \neq 0$ (so $\tan\alpha$ exists) and $1 - \tan^2\alpha \neq 0$. At $\alpha = 90°$, $\tan 2\alpha = \tan 180° = 0$ exists, but the right side is undefined, so use the sine and cosine formulas there.

Theorem

FROM THE HANDOUT

Double angle formulas are used to express the trigonometric ratios of double angles (2θ) in terms of trigonometric ratios of single angle (θ). The **Double Angle Formulas** of sine, cosine, and tangent are:

- $\sin(2\theta) = 2\sin\theta\cos\theta$
- $\cos(2\theta) = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1$
- $\tan(2\theta) = \frac{2\tan\theta}{1 - \tan^2\theta}$

Observation

FROM THE HANDOUT

You may have noticed that $\cos(2\theta)$ has three different forms. This is because we can use Pythagorean Identities to obtain the other forms. For example, suppose we start with $\cos(2\theta) = \cos^2\theta - \sin^2\theta$. We can substitute the first $\cos^2\theta$ with $1 - \sin^2\theta$ (think about how you can rewrite the first Pythagorean Identity, $\cos^2\theta + \sin^2\theta = 1$). We will then have $1 - \sin^2\theta - \sin^2\theta$, which simplifies to $1 - 2\sin^2\theta$. Thus, $\cos(2\theta) = 1 - 2\sin^2\theta$. Using the same method, you can get $2\cos^2\theta - 1$.

Activity 9

Activity 9 · from your notes

Suppose $\sin\alpha = \frac{2}{3}$ and α lies in Quadrant I. Use the Pythagorean Identities and Double Angle Formulas to find $\sin 2\alpha$.

Try it, then check.

Hint: $\sin 2\alpha = 2 \sin \alpha \cos \alpha$, so you need $\cos \alpha$ first. Which identity gets $\cos \alpha$ from $\sin \alpha$? Which sign?

Answer

Step 1, find cos α: $\cos^2\alpha = 1 - \sin^2\alpha = 1 - \frac{4}{9} = \frac{5}{9}$, so $\cos \alpha = \pm \frac{\sqrt{5}}{3}$. Quadrant I, so $\cos \alpha$ is positive: $\cos \alpha = \frac{\sqrt{5}}{3}$.

Step 2, double angle: $\sin 2\alpha = 2 \sin \alpha \cos \alpha = 2\left(\frac{2}{3}\right)\left(\frac{\sqrt{5}}{3}\right)$

$$\sin 2\alpha = \frac{4\sqrt{5}}{9}$$

Activity 10

Activity 10 · from your notes

The double-angle formulas can be used to derive the reduction formulas, which are formulas we can use to reduce the power of a given expression involving even powers of sine or cosine.

(a) Start with the double angle formula $\cos 2\theta = 1 - 2 \sin^2\theta$. Use your algebra skills to solve for $\sin^2\theta$.

Answer (a)

$2 \sin^2\theta = 1 - \cos 2\theta$

$$\sin^2\theta = \frac{1 - \cos 2\theta}{2}$$

(b) Given the second double angle formula, $\cos 2\theta = 2 \cos^2\theta - 1$, solve for $\cos^2\theta$.

Answer (b)

$2 \cos^2\theta = 1 + \cos 2\theta$

$$\cos^2\theta = \frac{1 + \cos 2\theta}{2}$$

(c) Begin with $\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta}$. Use the formulas you created in parts (a) and (b) to rewrite $\tan^2\theta$.

Answer (c)

$\tan^2\theta = \left(\frac{1 - \cos 2\theta}{2}\right) \div \left(\frac{1 + \cos 2\theta}{2}\right)$. The 2s cancel.

$$\tan^2\theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$$

Domain check: this needs $\cos\theta \neq 0$, which is the same as $1 + \cos 2\theta \neq 0$.

Theorem

FROM THE HANDOUT

The trigonometric power reduction formulas allow us to rewrite expressions involving trigonometric terms with trigonometric terms of smaller powers. The **Power Reduction Formulas** are:

- $\sin^2\theta = \frac{1 - \cos 2\theta}{2}$
- $\cos^2\theta = \frac{1 + \cos 2\theta}{2}$
- $\tan^2\theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$

Activity II

Activity II · from your notes

Rewrite $\sin^4\beta$ as an expression without powers greater than one.

(a) How can we rewrite $\sin^4\beta$ as a function squared?

Answer (a)

$$\sin^4\beta = (\sin^2\beta)^2$$

(b) Rewrite $\sin^2\beta$ using a Power Reduction Formula.

Answer (b)

$$\sin^4\beta = \left(\frac{1 - \cos 2\beta}{2}\right)^2$$

(c) Square the right-hand side of the equation you found in part (b).

Answer (c)

$$\sin^4\beta = \frac{1 - 2\cos 2\beta + \cos^2(2\beta)}{4}$$

(d) You still have a cosine squared. Substitute for it using a power reduction formula, then simplify.

Hint: Multiply the top and bottom of the inner fraction by 2 to clear it.

Answer (d)

The squared term is $\cos^2(2\beta)$, so the reduction formula doubles the angle again: $\cos^2(2\beta) = \frac{1 + \cos 4\beta}{2}$.

$$\sin^4\beta = \frac{1 - 2\cos 2\beta + \frac{1 + \cos 4\beta}{2}}{4} = \frac{2 - 4\cos 2\beta + 1 + \cos 4\beta}{8}$$

$$\sin^4\beta = \frac{3 - 4\cos 2\beta + \cos 4\beta}{8}$$

Your handout says "substitute the $\cos^2\beta$". The term being replaced is $\cos^2(2\beta)$.

Activity 12

Activity 12 · from your notes

The next set of identities is the set of half-angle formulas, which can be derived from the reduction formulas. Let's derive the half-angle formula for $\sin\frac{\theta}{2}$.

(a) Recall $\sin^2\theta = \frac{1 - \cos 2\theta}{2}$. Replace θ with $\frac{\theta}{2}$.

Answer (a)

$$\sin^2\left(\frac{\theta}{2}\right) = \frac{1 - \cos\left(2 \cdot \frac{\theta}{2}\right)}{2}$$

(b) Simplify the right-hand side of your equation in part (a).

Answer (b)

$2 \cdot \frac{\theta}{2} = \theta$

$$\sin^2\left(\frac{\theta}{2}\right) = \frac{1 - \cos\theta}{2}$$

(c) Now take the square root of both sides to solve for $\sin\left(\frac{\theta}{2}\right)$.

Answer (c)

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\left(\frac{1 - \cos\theta}{2}\right)}$$

The sign depends on which quadrant $\frac{\theta}{2}$ is in.

Your handout says "solve for sin θ". It should be $\sin\left(\frac{\theta}{2}\right)$.

Note

FROM THE HANDOUT

The derivation for $\cos\left(\frac{\theta}{2}\right)$ and $\tan\left(\frac{\theta}{2}\right)$ is similar to that in the previous example when starting with the power-reduction formulas for each trigonometric function.

Theorem

FROM THE HANDOUT

The **Half Angle Formulas** are (choose + or − depending on which quadrant $\frac{\theta}{2}$ is in):

- $\sin\frac{\theta}{2} = \pm\sqrt{\left(\frac{1 - \cos\theta}{2}\right)}$
- $\cos\frac{\theta}{2} = \pm\sqrt{\left(\frac{1 + \cos\theta}{2}\right)}$
- $\tan\frac{\theta}{2} = \pm\sqrt{\left(\frac{1 - \cos\theta}{1 + \cos\theta}\right)} = \frac{\sin\theta}{1 + \cos\theta} = \frac{1 - \cos\theta}{\sin\theta}$

Which sign?

✎ EXTRA NOTE: NOT IN YOUR HANDOUT

Rule: the sign comes from the quadrant of the **half angle**, $\frac{\theta}{2}$, not the original angle θ. (If your printed handout says "which quadrant θ is in,"

correct it to $\frac{\theta}{2}$.) For sin 15°, look at 15° (Quadrant I, so +), not 30°.

Activity 13

Activity 13 · from your notes

Find sin 15° using a half-angle formula.

(a) Which quadrant is the angle 15° in?

Answer (a)

Quadrant I

(b) What angle is 15° half of?

Answer (b)

30°

(c) Which half-angle formula should we use to determine sin 15°?

Answer (c)

15° is in Quadrant I, where sine is positive, so take the + root:

$$\sin\frac{\theta}{2} = \sqrt{\left(\frac{1-\cos\theta}{2}\right)}$$

(d) Substitute θ with 30° in the half-angle formula for sine.

Answer (d)

$$\sin\left(\frac{30°}{2}\right) = \sqrt{\left(\frac{1-\cos 30°}{2}\right)}$$

(e) Use the unit circle to find the exact value of cos 30° and substitute it into your equation from part (d).

Answer (e)

$$\sin 15° = \sqrt{\left(\frac{1-\frac{\sqrt{3}}{2}}{2}\right)}$$

(f) Simplify your equation in part (e) to find the exact value of $\sin 15°$.

Answer (f)

$1 - \frac{\sqrt{3}}{2} = \frac{2 - \sqrt{3}}{2}$, and dividing by 2 gives $\frac{2 - \sqrt{3}}{4}$.

$\sin 15° = \sqrt{\left(\frac{2 - \sqrt{3}}{4}\right)}$

$$\boxed{\sin 15° = \frac{\sqrt{(2 - \sqrt{3})}}{2}}$$

Check: ≈ 0.2588 ✓

Activity 14

Activity 14 · from your notes

Use the appropriate half-angle formula to find the exact value of $\cos 75°$.

Try it, then check.

Hint: Which angle is 75° half of? Which quadrant is 75° in?

Answer

$75° = \frac{150°}{2}$, and 75° is in Quadrant I, so cosine is **positive**.

$\cos 75° = \sqrt{\left(\frac{1 + \cos 150°}{2}\right)} = \sqrt{\left(\frac{1 - \frac{\sqrt{3}}{2}}{2}\right)} = \sqrt{\left(\frac{2 - \sqrt{3}}{4}\right)}$

$$\boxed{\cos 75° = \frac{\sqrt{(2 - \sqrt{3})}}{2}}$$

⚠ **Watch out:** $\cos 150°$ is negative, but that doesn't make $\cos 75°$ negative. The sign comes from where 75° is. The textbook's answer key shows a minus sign here; that's an error.

Compare Activity 5: there we got $\cos 75° = \frac{\sqrt{6} - \sqrt{2}}{4}$. Both are ≈ 0.2588. Same number, two different-looking exact forms.

Practice Example 15

Practice Example 15 · from your notes

Use a sum formula to find the exact value of $\cos(75°)$ by writing $75° = 45° + 30°$.

Try it, then check.

Answer

$\cos(45° + 30°) = \cos 45° \cos 30° - \sin 45° \sin 30° = \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$

$$\cos 75° = \frac{\sqrt{6} - \sqrt{2}}{4}$$

Practice Example 16

Practice Example 16 · from your notes

If $\sin(\theta) = \frac{3}{5}$ and θ is in the first quadrant, use a double-angle identity to find $\sin(2\theta)$.

Try it, then check.

Answer

$\cos\theta = \sqrt{\left(1 - \frac{9}{25}\right)} = \sqrt{\left(\frac{16}{25}\right)} = \frac{4}{5}$ (positive in Quadrant I)

$\sin 2\theta = 2 \sin\theta \cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right)$

$$\sin(2\theta) = \frac{24}{25}$$

Practice Example 17

Practice Example 17 · from your notes

Verify the identity $\frac{1 - \cos(2\theta)}{\sin(2\theta)} = \tan(\theta)$ by rewriting the left side using double-angle identities.

Try it, then check.

Hint: Which form of $\cos(2\theta)$ makes $1 - \cos(2\theta)$ simplest?

Answer

Use $\cos(2\theta) = 1 - 2\sin^2\theta$, so $1 - \cos(2\theta) = 2\sin^2\theta$. And $\sin(2\theta) = 2\sin\theta\cos\theta$.

$$\frac{1-\cos(2\theta)}{\sin(2\theta)} = \frac{2\sin^2\theta}{2\sin\theta\cos\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta \checkmark$$

Verified: the left side simplifies to tan θ

Domain check: the left side needs $\sin(2\theta) \neq 0$, that is, $\theta \neq k \cdot 90°$ ($\theta \neq k \cdot \pi/2$, where k is any integer). $\tan\theta$ has a larger domain: at $\theta = 0$, $\tan 0 = 0$, but the left side is $0/0$, undefined. The identity holds where both sides are defined.

Summary

FROM THE HANDOUT

Write 2–3 sentences summarizing the key concepts of today's lecture.

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